

Science outreach activities in Kashmir-Leh-Ladakh

Summary: This document contains a report of science outreach activities conducted by a group of volunteers from Pune, closely associated with Disha, a voluntary organization of students of Indian Institute of Science Education & Research (IISER) – Pune and Jnana Prabodhini, Pune. The group of six volunteers travelled across Kashmir-Leh-Ladakh area from 7th May to 22nd May, 2013 to conduct science workshops for students in 12 schools in remote areas. The organizations that supported us to get connected with the schools were, Aseem Foundation and Nature Conservation Foundation (NCF). The workshops that would typically run for 3-4 hours in a school for 5th to 8th standard students were designed, to convey joy and excitement in science through science demonstrations, discussions, hands-on experiments; and to encourage students to develop interest in scientific thinking.

A group of six volunteers closely associated with Disha, of IISER Pune and Jnana Prabodhini, Pune, travelled across Kashmir-Leh-Ladakh area from 7th May to 22nd May, 2013 to conduct science workshops. Four IISER-Pune students – Nishad Mandlik, Mihir Umarani, Suhas Gawali and Kaustubh Deshpande; a Pune based IT professional- Kalyani Katariya; and a freelance consultant- Sugat Dabholkar participated in the tour.

The volunteers of the group have been involved in various science education activities over past few years, conducting science camps and workshops in various remote parts of India including North-East India, Chhattisgarh along with slums and rural areas nearby Pune.

In order to extend our efforts to bring joy and excitement in science to school students, we conducted science education workshops at schools in Kashmir – Leh -Ladakh area, during our 15 days visit there in May 2013. Being very enthusiastic and passionate about bringing education (and particularly science-based education) to all under-served children, our focus of this visit was primarily on science education activities involving hands-on science experiments, toys, games, competitions etc. at these schools.

We would conduct workshops for around 2 to 5 hours, depending on how much time the schools would allow us to use. We had designed the workshops for 5th to 8th standard students, but slightly younger or older students were also welcome to join in. The workshops included a story, songs, science demonstrations, discussions, hands-on experiments for students. We used very simple stuff (not economically fancy, but intellectually exciting) for the demonstrations and experiments largely based on Arvind Gupta's toys (<http://www.arvindguptatoys.com/toys.html>), so that students can relate to what we're talking about, get excited about science and develop interest in scientific way of thinking.

Brief description of workshops

1. Khushhaal educational trust, Ganderbal:

After a train journey from Pune to Jammu followed by overnight road travel, we reached Srinagar on 8th May. After a bit of sight-seeing in the beautiful city, we reached Ganderbal where we stayed in the homes of school students there.

Our first science workshop of the tour was conducted at Khushhaal educational trust school here the next day on 9th May, attended by 70-75 students (including students from nearby New Mission school). Assem foundation, a Pune-based organization working for social issues in Jammu-Kashmir and also a few other regions of India, helped us get connected with this school to conduct the workshop. Mehraj Ue Din Lone, principal of the school, and also all other teachers were very cooperative. We enjoyed the workshop as well as the hospitality at the homes where we stayed for a day.



2. Govt. middle school, housing colony, Leh:

After a long journey from Srinagar to Leh, enjoying the scenic road via Drass and Kargil, we reached the city in the night of 10th May. We were very happy to receive the warm welcome by Karmaji Sonam (NCF field coordinator at Leh) and Namgyal family (owners of the guest house).



The first workshop in Leh-Ladakh area was conducted the very next day, 11th May, at Govt. middle school in housing colony, Leh. Close to 40-45 students of 5th-10th standard from the school participated in the workshop.

3. Visit to SECMOL, Phey campus:

Students' Educational and Cultural Movement Of Ladakh (SECMOL) is a very renowned organization working in the field of education in Ladakh. Founded by Mr. Sonam Wangchuk in 1980s, the organization worked for many years at government schools to improve quality of education, even in remote areas of Ladakh. We visited their campus at Phey village (around 20 km from Leh) where we met Rebecca Norman, wife of Mr. Sonam Wangchuk, who showed and explained to us various activities running there. The campus houses around 40 students of the age group 18-23. These are mostly drop-out students who, though not having any formal course here, learn practical knowledge regarding agriculture, energy conservation, and waste management along with classes on regular academics conducted by some regular teachers and mostly outside volunteers. SECMOL seems to have a very strong volunteer programme through which many Indian and even interested foreign people visit the place to devote their time and expertise for the students. At the time we visited the

campus, 5 Teach for India fellows were spending their vacation as volunteers there. Though we could not conduct our workshop here, the visit was certainly fruitful as we came to know the work of the organization.

4. Bharatiya Vidyaniketan school, Choglamsar:

This is a residential school, run in close association with Vanvasi Kalyan Ashram. Principal Gopalji Thakur and a few other teachers were very cooperative. We conducted science workshop at this school on 13th May. Around 50 students enthusiastically participated in this 4 hour-long workshop.

5. Govt. primary school, Ayu, Saboo:

On 14th May, we conducted the workshop at Saboo, a small village situated near Leh. Separate workshops were conducted for students and teachers, in parallel. Since the kids



were younger (1st – 4th standard), the workshop for them included primarily playing with the scientific toys and also making a few of them. Mr. Lobzang Angdus, principal of primary school, was very proactive towards arranging the workshop, especially for the teachers. The teachers (around 10 in number) were very enthusiastic and also inquisitive. We showed them all the toys/experiments in our kit and also explained how these can be used to teach science in a better and more

interesting way. This seemed to appeal to them as they tried to make as many of them, using the extra material we had, to use for their students.

6. Centralized govt. residential school, Tharuk:

This school is situated in Tharuk which is a small village near Tangtse (another village near Pangong lake). After our visit to Pangong lake, we conducted the workshop here on 15th May, for around 30-35 students. Situated in very beautiful surroundings, the school had equally vibrant and innovative kids who participated in the workshop with great energy.



7. Govt. high school and White Lotus school, Shey:

On 16th May, we conducted workshops simultaneously at Govt. high school and White Lotus school, both situated in Shey. At the govt. high school, 25-30 students participated in the workshop. Mrs. Sonam Angmo, principal of the school, was very helpful. We could conduct the workshop for almost 5 hours. The kids were very excited and participated with



great enthusiasm. We also tried to conduct “Idea competition” for the kids. Though the students may not have completely understood the competition, we certainly got a brief look inside their thinking about their lives.

White Lotus school and its principal Mr. Prasad Eledath too were highly receptive to us. Considering the high number of total students, we conducted the workshop here only for 8th standard students (around 40).

8. Govt. middle school, Khardung:

The next day (17th May) we set out for 2-day visit to Nubra valley and the schools there. Just after passing Khardungla, world’s highest motorable road, we came to the village Khardung. Here we conducted science workshops at two government schools situated in different parts of the village (lower Khardung and upper Khardung or Resermo). Both these schools had very low population of students, 10 in one and 15 in the other. The teachers were very interested in our workshop, participated with the kids and also helped us communicate with them. The kids, though a bit shy and slightly less familiar with Hindi, were very cute, playful (especially the younger ones!) and tried to learn most of the things. The workshop was indeed enjoying experience for them as it was for us.



9. Govt. tribal hostel, Diskit:

We reached Diskit, the major town of Nubra valley, on 17th evening immediately after which we approached and conducted the workshop at govt. tribal hostel there. A teacher in one of the schools at Khardung had suggested us this place. Though the workshop could be conducted only for around 2 hours here (due to time constraints), the students (around 70 in number and of the age group 7th – 10th standard) enjoyed most when they did the experiments on their own.

10. Govt. middle schools in Skampuk and Skuru:

The next day (18th May), after visiting the desert (and sand dunes, double-hump camels) nearby in Hunder, we conducted our workshop simultaneously at two nearby govt. middle schools, in villages Skampuk and Skuru. Skampuk is situated adjacent to Hunder while Skuru 15 km from it, near the airbase Thoise (last base of Indian air





force towards Siachen border). Both these village schools had a small number of students (10-15), including the younger ones. Many of them were very eager when it came to do things by hands. It was really pleasing to see some of them, despite living in such remote area, really having the aptitude towards science.

Thus, we conducted a total of 12 science workshops, 1 in Ganderbal and 11 in Leh-Ladakh area. As mentioned earlier, both students and teachers were very eager and curious about the workshop in all the schools we visited. The kids enjoyed doing hand-on experiments/toys and also paper caps the most. The kids were inquisitive and tried to ask us questions (like why is there less rainfall in Ladakh etc.). Some of them were really thoughtful and also innovative in giving answers through simple experiences (e.g. relation of pressure, force and area using blowing of air by mouth). Living in very beautiful surroundings, they were very imaginative which was evident in their answers. e.g. When asked how the wave model (made up of straws and tape) looks like, they would answer mountains, leaves, fish, ladder as against the bookish answers given by the city students like DNA or wave itself!

We visited primarily the government schools situated in rural areas. Though these were provided with decent infrastructure and other facilities including qualified teachers, the population at these schools was very less. The main reason for this, as mentioned by the teachers there, is the recent trend - even in remote villages - to send the kids to private schools in nearby cities (mainly Leh) due to the polished outlook (!) of the students there. Only the children of families with low economic background and migrant labourers (from Nepal, Bihar, Tibet) study in government schools. We experienced this in some of the schools.

Along with our activities at the schools, we could also briefly experience and understand the local life and culture through our stay at the homes of people there. All our activities were possible only due to the unconditional help and support provided to us by the local people - Mehraj sir in Ganderbal; Karmaji Sonam and Dorji in Ladakh. We would like to express our deep gratitude towards them and also all others, individuals and organizations, who helped us in organizing this tour.

[Report compiled by: Kaustubh Deshpande, Kalyani Katariya and Sugat Dabholkar]